

Epistemic Transition Integrity

Foundations of Correction-Sufficient State
and Sequential Revision

Zackary Loevseth

Research direction and synthesis with substantial AI assistance

7 September 2026

ETI-SYNTHESIS-2026-09-07.1

Abstract

Long-running research must respond to changed evidence without retaining stale conclusions or discarding work that remains justified. An answer summary alone may omit the distinctions needed for that response. Epistemic Transition Integrity (ETI) studies the preservation and use of task-relevant evidence, support, authority, and correction state. This paper synthesizes a conceptual foundation, a finite mathematical kernel, a stochastic repair model, and a sequential extension in one explicit account. For compatible edits whose acceptable sets are Boolean intervals, a summary permits a common correction exactly when the union of required edits is contained in the intersection of permitted edits; minimum one-shot representations are graph colorings. Under unrestricted additive stochastic repair, Bayes-optimal action sets have that same interval form for fixed costs, while preserving optimality for all positive costs requires the posterior-impact signature under common signal support. Online maintenance of a fixed encoding of the current reference state additionally requires transition congruence. We prove the characterization, smallest strict memory gap, limits of coarsest quotients, and a finite proposed-merge test. Grounded alternative support supplies a concrete operational example throughout. The guarantees are conditional on explicit models and correctly constructed state; they do not establish empirical human–AI superiority. The resulting contribution is a consolidated specification, connected proofs, counterexamples, reproducible checks, and a falsifiable evaluation program using established mathematical machinery.

Keywords: epistemic state; correction; bounded representation; provenance; decision theory; finite-state abstraction; human–AI research.

Contents

1	Research problem and scope of the synthesis	3
2	An inspectable epistemic architecture	4
3	Finite correction under partial information	6
4	Grounded support and exact reopening	9
5	Stochastic repair and the value of retained information	10
6	Maintaining a correction-sufficient state online	14
7	Minimum size, coarsest quotients, and proposed merges	15
8	Relation to established work	17
9	Evidence and reproducibility	18
10	A falsifiable empirical program	20
11	Contribution, limitations, and accountability	21
A	A fixed scalar closure path	22
B	Finite algorithms and explicit input costs	23
C	Source lineage and corrected statements	24
D	Regimes at a glance	25

1 Research problem and scope of the synthesis

Research is a sequence of commitments made under incomplete information. A source is admitted, an interpretation supports a claim, a claim supports a later decision, and a result is eventually described as complete. New evidence can challenge any of these steps. A useful correction changes the commitments that are no longer justified while preserving those that still have adequate support. This is difficult when a system retains the final conclusion but loses the reasons, conditions, or authority on which it depended.

Epistemic Transition Integrity (ETI) names a research program about this continuity of justification. Its central question is:

What distinctions must a bounded research process preserve so that later evidence can produce a justified correction, and how can those distinctions be maintained through successive changes?

The word *integrity* refers to consistency between the stated task, the available support, the permitted transition, and the recorded outcome. It is not a guarantee that a belief is true, that a source is trustworthy, or that an implementation is reliable. Those require separate evidence. The program's broad foundation also concerns the role of human judgment, metacognition, and reciprocal criticism in identifying what a representation has omitted [1]. The finite mathematics below isolates one part of that problem: information needed for specified correction decisions.

1.1 Why a summary may be insufficient

Two research histories can produce the same sentence, “the conclusion is supported,” while carrying different correction obligations. In one history the conclusion has two independent evidential supports. In the other it has only one. Withdrawal of their common support requires different responses. Preserving the sentence accurately does not necessarily preserve the ability to revise it accurately. The relevant object is therefore a task-relative *decision distinction*, rather than a word count or a general measure of semantic similarity.

This observation does not imply that longer context, a database, retrieval, or a structured ledger will always improve performance. A record can contain the wrong dependency, misidentify a source version, or claim a permission that was never granted. Constructing, checking, retrieving, and updating the record also have costs. The research question is whether a particular representation preserves the distinctions needed for a declared task, and whether an actual system can construct and use it better than an adequately resourced alternative.

1.2 Human–AI correction as a research hypothesis

The broad foundation proposes reciprocal blind-spot correction: people and AI systems may help expose omissions in one another's representation of a task. A person can question the target, relevance, authority, or practical meaning of an output; an AI system can help enumerate cases, expose inconsistent commitments, or reconstruct an argument for inspection. Which contribution is useful depends on the task, available evidence, tools, and competence. Neither role has automatic epistemic authority.

Here a *task-relative blind spot* is an omitted, inaccessible, or misweighted distinction whose recovery could change a warranted decision. This is an operational description. It does not identify every

psychological bias with lost information, establish that an agent is unable to assess its own error, or characterize consciousness. Studies of bias perception and metacognitive inference help motivate careful separation of judgment and confidence [6, 7]; they do not prove the finite representation theorems in this paper. Conversely, those theorems do not explain lived experience or demonstrate a general theory of intelligence.

Human–AI collaboration has to be evaluated against the stronger relevant baseline. The existence of a useful human contribution and a useful AI contribution does not by itself establish that their combination exceeds either component at a comparable cost. Task-dependent findings on combined performance and overreliance motivate an explicit empirical comparison [8, 9]. The present paper provides a specification and bounded mathematical results for such a comparison, rather than a measured performance advantage.

1.3 What is combined here

This edition reconciles five local research documents: the September 4 foundational reference, its mathematical kernel, the September 7 concise finite note, the August 13 RR-ETI v2 decision-theoretic manuscript, and the September 7 sequential-state note [1, 2, 4, 3, 5]. Their source identities and treatment are recorded in Appendix C. The concise finite note and kernel share mathematical content; they are not counted as independent confirmations. Later corrections govern the present exposition. Historical execution and external review claims are not inferred from a rewritten manuscript.

The contribution is a unified correction-specific account with three mathematical regimes:

1. **Robust feasibility.** A common correction must be acceptable in every reference state compatible with the retained information.
2. **Decision-theoretic optimality.** A repair minimizes expected loss under a specified probability model and costs. The entire set of optimal repairs, including ties, matters.
3. **Sequential maintainability.** A fixed encoding of the current reference state must be updated from its retained symbol and each observed event, under an explicit transition model.

The same interval algebra connects the first two regimes in the unconstrained additive case. Transition congruence supplies the extra requirement in the third. The mathematical tools are established: set intersection, graph coloring, finite fixed points, conditional expectation, and deterministic state minimization. The formulation, integration, and worked boundary cases are the research object; independent priority for the component mechanisms is not claimed.

2 An inspectable epistemic architecture

The broad architecture describes a research process more general than any one finite model in this paper. Let Σ_t denote its recorded state at version t . It can contain a task specification, source records, claim records, admitted support rules, authority conditions, proposed actions, receipts, and validation results. These are typed records with distinct meanings. Their existence does not certify their accuracy.

Record	Meaning and necessary distinction
Task contract	Target, scope, permitted evidence, output requirements, and completion criteria. A revised target is a new version.
Evidence	A source, its version, admissibility, and relevant content. Evidence can support a claim without authorizing an action.
Claim and assertion	A proposition is distinct from an event in which someone or something asserted it. Repetition is not independent support.
Support rule	Which premises jointly support a conclusion, and which derivations provide alternatives. Admission and validity are explicit assumptions.
Authority and capability	Permission in a scope and version is distinct from the technical ability to act. Both may be necessary.
Action and receipt	Proposal, attempted execution, and observed outcome are separate events. A planned result is not an execution receipt.
Validation and closure	A specified check at a specified version, and a completion judgment against the task's current contract.

2.1 Support, authority, and closure

A well-typed record prevents some category errors but supplies no oracle for admissibility. Source trust can depend on methodology, access, provenance, and context. A support rule can be a formal inference, a defeasible scientific argument, or a policy-dependent association. The positive-Horn illustration in Section 4 addresses only the first, restricted case. Inconsistency, defeasible exceptions, and uncertain rule admission require additional semantics.

An operational action gate can be specified as a conjunction of current permission, capability, and task-specific prerequisites. This is an engineering contract, not a derivation of legitimate authority. The mechanisms that obtain and validate those inputs remain part of the system's burden. A stale authorization record cannot become valid merely because it is present in memory.

Completion is similarly version-relative. It requires the declared output, the applicable checks, and any necessary outcome evidence for the current target and source state. A correction can invalidate a closure judgment while leaving some underlying propositions true. A task can be paused or blocked without being completed. The architecture therefore records why a status changed and which source versions it concerns, while preserving earlier assertion and validation events as history.

2.2 A sustained support-withdrawal example

Consider two independently admitted evidence roots a, b , a claim c , and a downstream claim d . The fixed rules are

$$a \Rightarrow c, \quad b \Rightarrow c, \quad c \Rightarrow d.$$

One reference environment contains both roots; another contains only a . Before a withdrawal, both environments justify the same recorded status: c and d have grounded support. An event withdraws a . If b remains admitted and available, both claims remain supported. If it does not, both lose all grounded support.

Take the edit atoms to be r_c and r_d , meaning “reopen the support-status record for this claim.” For the illustrative contract *reopen exactly the claims that lose grounded support*, the required and permitted patch is $\emptyset$ in the first environment and $\{r_c, r_d\}$ in the second. A representation storing only the two initial supported-status bits has merged environments needing disjoint corrections.

Recovering whether b is available and admitted can resolve that ambiguity; rephrasing the same two status bits cannot.

This example distinguishes reopening from refutation. Withdrawal of the last admitted proof of c does not prove $\neg c$. It also distinguishes support from authority: a justified claim can still be used in an action that requires an additional permission. The example stipulates the exact reopening contract for clarity. Real tasks may permit optional checks or require other edits; their must/may sets must express those choices explicitly.

2.3 From recorded state to a formal model

The models below use a finite *reference state* x , not the entire operational record Σ_t . A reference state specifies the task-relevant distinctions and correct decision data for the model. It may describe the external evidence environment in the example. A stored representation of it need not be accurate in practice. In particular, a language model's estimate of a reference state is not identified with that state by definition.

For each theorem, the true reference state must lie in the retained information cell, the supplied task constraints must be correct, and the specified decoder or updater must actually be executed. Observing the raw material, admitting sources, constructing dependencies, and identifying permissions are upstream tasks. End-to-end reliability requires them to succeed as well. This separation allows a mathematical guarantee to be precise without assigning a guarantee to a fallible state constructor.

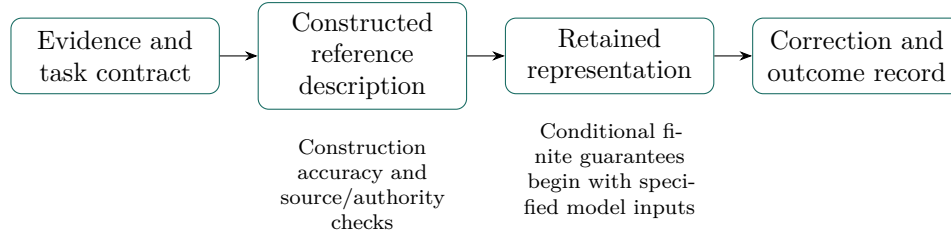


Figure 1: A representation theorem covers a declared part of the workflow. Construction, permission, execution, and outcome validation retain their own obligations.

3 Finite correction under partial information

Let X be a finite nonempty set of reference states, $\mathcal{E}$ a finite nonempty event set, and $\mathcal{V}$ a finite universe of mutually compatible elementary edits. Each event is defined at every reference state. For each (x, e) , supplied sets satisfy

$$M_e(x) \subseteq U_e(x) \subseteq \mathcal{V}. \quad (1)$$

The required set $M_e(x)$ specifies what must change, and $U_e(x)$ specifies what may change. The *complete interval assumption* is

$$\mathcal{A}_e(x) = [M_e(x), U_e(x)] := \{\Delta \subseteq \mathcal{V} : M_e(x) \subseteq \Delta \subseteq U_e(x)\}. \quad (2)$$

Thus every intermediate set is acceptable. Mutually exclusive actions, joint budgets, and logical constraints between edits may violate this assumption. Merely specifying lower and upper bounds does not establish that they describe the entire admissible family.

A deterministic encoder $c : X \rightarrow Z$ is chosen before the event. A decoder $d : Z \times \mathcal{E} \rightarrow 2^{\mathcal{V}}$ receives the retained symbol and the observed event. The full codebook and model are shared. For each used symbol z , write $X_z = c^{-1}(z)$; used cells are nonempty. The one-shot task requires

$$d(c(x), e) \in \mathcal{A}_e(x) \quad \text{for every } x \in X, e \in \mathcal{E}.$$

There is no distributional or average-case relaxation in this requirement.

3.1 The common-correction criterion

Define

$$M_e^+(z) = \bigcup_{x \in X_z} M_e(x), \quad U_e^-(z) = \bigcap_{x \in X_z} U_e(x), \quad B_e(z) = M_e^+(z) \setminus U_e^-(z).$$

The set $B_e(z)$ contains edits that some possible state requires and another forbids. It is a finite correction-conflict certificate, not a general psychological measure.

Theorem 3.1 (Common correction and the least patch). *A patch acceptable for every $x \in X_z$ exists if and only if $M_e^+(z) \subseteq U_e^-(z)$. When this holds,*

$$\bigcap_{x \in X_z} \mathcal{A}_e(x) = [M_e^+(z), U_e^-(z)],$$

and $M_e^+(z)$ is the unique inclusion-least common patch.

Proof. A common patch contains every $M_e(x)$ and therefore their union. It is contained in every $U_e(x)$ and therefore their intersection. Conversely, every set between this union and intersection satisfies each state's interval constraints. The union itself is such a set exactly when the stated inclusion holds, and it is contained in every common patch. $\square$

For any inclusion-monotone cost C on patches, the least patch minimizes C among the common patches. It need not be the unique scalar-cost minimizer: an optional extra edit can have zero incremental cost. Nor does a common patch have to be the cheapest patch at each fully known state. Robust feasibility and statewise cost optimality are distinct tasks.

Theorem 3.1 is an elementary intersection identity, already explicit in the kernel [2]. Its use is to make the missing-decision-information obstruction inspectable. If a fiber has $B_e(z) \neq \emptyset$, changing the decoder cannot solve that task from the same symbol and event. A legitimate additional observation can refine the fiber; more computation on identical information cannot separate its members.

Example 3.2 (The support summary loses a necessary distinction). In Section 2.2, the two environments have the same initial supported-status summary. After withdrawal of a , their acceptable families are the distinct singletons $\{\emptyset\}$ and $\{\{r_c, r_d\}\}$. Consequently $M_e^+ = \{r_c, r_d\}$ and $U_e^- = \emptyset$. The merged cell has no common correction. A representation distinguishing the availability of b suffices for this one event and contract.

3.2 Refinement and randomization at a conflict

For any nonempty state set B , define the conflict certificate

$$\mathcal{B}_e(B) = \left(\bigcup_{x \in B} M_e(x) \right) \setminus \left(\bigcap_{x \in B} U_e(x) \right).$$

Proposition 3.3 (Information refinement and a two-state lower bound). *If $\emptyset \neq B' \subseteq B$, then $\mathcal{B}_e(B') \subseteq \mathcal{B}_e(B)$. If an edit is required at x and forbidden at y , a randomized decoder receiving the same code and event at these two states has worst-case failure probability at least $1/2$. Its average failure is at least $1/2$ under the equal prior on these two witness states.*

Proof. Restricting the state set decreases the required union and increases the permitted intersection, proving the first inclusion. For the second claim, let p be the probability that the decoder includes the witnessing edit. At x its failure probability is at least $1 - p$, and at y it is at least p . Their maximum and their equal-weight mean have the stated lower bounds. The same output distribution is used because the decoder has the same observation and no additional state-correlated information. $\square$

The refinement is an actual reduction of possibilities within the fixed reference model. An unjustified exclusion of the true state does not gain this guarantee. The randomized lower bound concerns a declared two-state witness distribution; it is not an error-rate claim for an arbitrary workload. It shows why random choices cannot supply the missing distinction while preserving zero-error correction.

3.3 Exact one-shot representation size

Define the undirected correction-conflict graph G on X . Distinct states x, y are adjacent when, for some $e \in \mathcal{E}$,

$$M_e(x) \not\subseteq U_e(y) \quad \text{or} \quad M_e(y) \not\subseteq U_e(x).$$

There are no self-conflicts because of (1).

Theorem 3.4 (Interval compatibility and coloring). *An encoder admits a valid one-shot decoder if and only if each of its cells is an independent set of G . The minimum number of occupied symbols is $\chi(G)$. With a shared codebook, the minimum fixed-length binary message length is $\lceil \log_2 \chi(G) \rceil$ bits.*

Proof. If a cell contains an edge, its witnessing event has an edit required at one member and forbidden at another, so it cannot admit a common patch. Conversely, in an independent cell B , for every event and every $x, y \in B$ we have $M_e(x) \subseteq U_e(y)$. The same inclusion holds for $x = y$. Taking the union over x and intersection over y yields Theorem 3.1. Feasible cells are therefore exactly the color classes of proper colorings; minimizing their number gives $\chi(G)$. Encoding that many labels requires and admits the stated fixed number of bits. $\square$

This counts the retained message only. Codebook size, state acquisition, decoding, support construction, and sequential updating are outside the count. The result characterizes an optimum, without supplying a scalable coloring algorithm.

Example 3.5 (A finite compression example). Let $X = \{0, 1, 2, 3, 4\}$ and $\mathcal{V} = \{d_0, \dots, d_4\}$, with one event and indices modulo five. Set $M(i) = \{d_i\}$ and $U(i) = \mathcal{V} \setminus \{d_{i-1}\}$. The conflict graph is a five-cycle. The labeling $(0, 1, 0, 1, 2)$ supplies three valid cells, and the odd cycle cannot use two colors. Thus two message bits suffice, compared with three for the five exact identities, under the stated shared-codebook accounting. This is a model example, not a measured reduction in an AI system's end-to-end cost.

3.4 Why arbitrary acceptable actions need more than a graph

If acceptable families are general sets $\mathcal{A}_e(x)$, common correction is still exactly the condition $\bigcap_{x \in B} \mathcal{A}_e(x) \neq \emptyset$. The pairwise graph shortcut can fail. Let P, Q, R be three distinct possible patches and give three states the acceptable families $\{P, Q\}$, $\{Q, R\}$, and $\{P, R\}$. Every pair has an acceptable patch in common, while all three do not. A pairwise compatibility graph would permit an invalid merged cell. Higher-order conflict information or the original family-intersection condition is then necessary.

4 Grounded support and exact reopening

Let H be a finite set of evidence-root identifiers and C a finite set of claim identifiers, with $H \cap C = \emptyset$. The available admitted roots are $F \subseteq H$. Fix a finite set $\mathcal{R}$ of positive-Horn rules $P \Rightarrow c$, where $P \subseteq H \cup C$ and $c \in C$. Premises in a rule are conjunctive; different rules for a head provide alternatives. Starting from $J_0 = \emptyset$, define

$$J_{k+1} = J_k \cup \{c : \exists (P \Rightarrow c) \in \mathcal{R}, P \subseteq F \cup J_k\}, \quad J^*(F) = \bigcup_{k \geq 0} J_k. \quad (3)$$

An admitted empty-premise rule is an explicit axiom. It is not a license to treat an arbitrary unsupported claim as grounded.

Proposition 4.1 (Least grounded closure). *The iteration reaches a fixed point after at most $|C|$ strict expansions. Its result is the least rule-closed claim set containing the consequences of F , and consists exactly of claims with finite derivations grounded in F and admitted axioms. For fixed rules, $F' \subseteq F$ implies $J^*(F') \subseteq J^*(F)$.*

Proof. Each strict expansion adds a member of finite C . Every claim added at iteration $k+1$ has a rule whose premises are roots or earlier claims, giving a finite grounded derivation by induction. Conversely, induction on finite derivation depth places every grounded consequence in the iteration. Any rule-closed set must contain every iteration by induction, proving leastness. The same induction with $F' \subseteq F$ proves monotonicity. Stabilization follows after at most $|C|$ additions of nonempty layers. $\square$

If evidence is withdrawn from F to F' , the exact loss-of-support set is

$$D(F, F') = J^*(F) \setminus J^*(F').$$

For the contract of reopening exactly these records, assign an edit r_c to each claim and set $M = U = \{r_c : c \in D(F, F')\}$. This is a direct instance of the finite interval model. A wider task can permit further edits, but must state that explicitly.

In the running example, removal of a leaves c, d grounded if b remains available. Reopening every graph descendant of a would therefore do unnecessary work. Conversely, rules $c \Rightarrow d$ and $d \Rightarrow c$ without a grounded entry derive neither claim. An algorithm that treats mutual reference alone as support violates (3). These two cases expose different errors: overlooking alternatives and accepting ungrounded cycles.

The result is limited to fixed admitted positive rules. In this setting the intersection of all closed claim sets is also the least closure, giving a separate finite oracle for implementation checking. Truth-maintenance systems already represent assumption-dependent support and alternatives [10]; provenance semirings supply substantially richer derivational accounts [11]. ETI's use of the small Horn

model does not claim either mechanism as an invention. Its purpose here is to supply explicit, checkable correction obligations.

4.1 The same example across successive withdrawals

The support example also makes the maintenance problem concrete. Let the reference state be the current root set

$$X = \{\emptyset, \{a\}, \{b\}, \{a, b\}\},$$

and let events e_a, e_b remove the named root by set difference, even if it is already absent. The output is the exact reopening patch $\{r_c, r_d\}$ when the event removes the last available root, and $\emptyset$ otherwise. These are actions on the claim records; the reference transition changes the external evidence-root set and does not depend on that action.

The current output tables of $\emptyset$ and $\{a, b\}$ are identical: neither individual withdrawal causes a new support loss. They can share a one-shot code. States $\{a\}$ and $\{b\}$ each have a different output table, so exactly three one-shot codes suffice. But after e_a , the two merged states become $\emptyset$ and $\{b\}$, which require different outputs on e_b . A representation of the current reference state that can be maintained under all event sequences must therefore distinguish all four states. Section 6 proves the general condition underlying this three-versus-four example.

5 Stochastic repair and the value of retained information

RR-ETI asks which retained information preserves a good repair decision when the correction signal is noisy and mistakes have different costs [3]. This is a finite decision problem. Its probability model and loss replace neither the authority conditions nor the robust validity requirements of Section 3.

5.1 Probability model, actions, and loss

Let W take values in a finite nonempty context set Ω , with prior $\mu(w) > 0$. Conditional on $W = w$, a latent event D has a finite distribution π_w , and an observed signal Y has conditional law $K_w(y \mid D)$ on a finite nonempty alphabet $\mathcal{Y}$. The impact indicator $I_i(w, D) \in \{0, 1\}$ says whether artifact $i \in \mathcal{I} = \{1, \dots, m\}$ is affected. A null event may be included and has zero impacts. For the results stated over every context and signal, assume *common positive signal support*:

$$p_w(y) := \mathbb{P}(Y = y \mid W = w) > 0 \quad \text{for all } w, y.$$

An encoder $t : \Omega \rightarrow Z$ is fixed before observing Y . The decoder receives $t(W)$, Y , and the declared costs, and selects a subset $S \subseteq \mathcal{I}$ to reopen. Initially all subsets are feasible. Let $s_i > 0$ be the penalty for leaving an affected artifact active and $r_i > 0$ the penalty for reopening an unaffected one. These costs are common across contexts. The loss is

$$L_w(S, D) = \sum_i [s_i I_i(w, D) \mathbf{1}_{i \notin S} + r_i (1 - I_i(w, D)) \mathbf{1}_{i \in S}]. \quad (4)$$

Correct reopening is assigned zero loss in this classification-style model. It is not a complete model of execution time, validation expense, or irreversible side effects. No independence between artifact impacts is assumed; additivity and the unrestricted action family are what permit coordinatewise decisions.

Define posterior impacts and their compressed counterparts by

$$q_i(w, y) = \mathbb{P}(I_i(W, D) = 1 \mid W = w, Y = y), \quad (5)$$

$$\bar{q}_i(z, y) = \mathbb{P}(I_i(W, D) = 1 \mid t(W) = z, Y = y) = \mathbb{E}[q_i(W, y) \mid t(W) = z, Y = y]. \quad (6)$$

The last identity is the finite tower property. On a fiber $t^{-1}(z)$ the averaging weight of context w is proportional to $\mu(w)p_w(y)$. Averaging only by the prior while ignoring the signal likelihood can give the wrong compressed posterior.

5.2 Bayes optima form correction intervals

Set $\theta_i = r_i/(s_i + r_i)$ and define

$$M_y(w) = \{i : q_i(w, y) > \theta_i\}, \quad U_y(w) = \{i : q_i(w, y) \geq \theta_i\}.$$

Theorem 5.1 (Complete optimal-action interval). *The entire set of Bayes-optimal repairs conditional on (w, y) is*

$$\text{Opt}(w, y) = [M_y(w), U_y(w)].$$

For fixed costs, one repair is optimal at every context in a nonempty retained cell B exactly when

$$\bigcup_{w \in B} M_y(w) \subseteq \bigcap_{w \in B} U_y(w).$$

The minimum one-shot code count preserving contextwise optimality at every signal is the chromatic number of the graph joining two contexts whenever some signal and coordinate give opposing strict preferences.

Proof. Conditional on (w, y) , leaving artifact i active costs $s_i q_i$, and reopening it costs $r_i(1 - q_i)$. Reopening is strictly preferable exactly when $q_i > \theta_i$, retaining is strictly preferable exactly when $q_i < \theta_i$, and both are optimal at equality. The unrestricted additive action space permits each tied choice independently. Hence all and only the sets in the stated interval are optimal. The remaining statements are Theorems 3.1 and 3.4 applied to these intervals. A conflict requires $q_i(w, y) > \theta_i > q_i(w', y)$, or the reverse. $\square$

This is the precise link between the robust and stochastic branches. In the former, must/may sets are stipulated validity constraints. In the latter they are derived optimality constraints under a probability model and fixed costs. The mathematical interval identity applies to both, while the meaning of success differs.

For example, one artifact with impact probability $1/4$ and equal costs is optimally retained, accepting expected loss $1/4$. If a separate robust task requires exact reopening when the latent impact occurred and both outcomes remain possible, neither action meets that task in both worlds. The Bayes rule is still optimal for (4). A task can instead minimize expected loss subject to a robust feasible family, but it must specify that additional constraint.

5.3 Exact representation regret

For fixed costs define the concave piecewise-affine function

$$g_i(q) = \min\{s_i q, r_i(1 - q)\}.$$

Let B_{full} be minimum expected loss for information (W, Y) and B_t for $(t(W), Y)$, using the same loss and action space. All sums over retained symbols use the image $t(\Omega)$, so every conditioning pair (z, y) has positive probability. Finite conditional minimization gives

$$B_{\text{full}} = \mathbb{E} \sum_i g_i(q_i(W, Y)), \quad B_t = \mathbb{E} \sum_i g_i(\bar{q}_i(t(W), Y)).$$

Theorem 5.2 (Jensen representation gap, including ties). *Writing $Z = t(W)$,*

$$B_t - B_{\text{full}} = \sum_{z,y} \mathbb{P}(Z = z, Y = y) \sum_i \left[g_i(\mathbb{E}[q_i(W, y) \mid z, y]) - \mathbb{E}[g_i(q_i(W, y)) \mid z, y] \right] \geq 0. \quad (7)$$

Under the positive joint-support assumptions, equality holds if and only if no fiber, signal, and co-ordinate contains posterior values strictly on opposite sides of θ_i . Equivalently, every cell admits a common contextwise Bayes-optimal repair at every signal.

Proof. Subtract the two conditional risk formulas and use (6) to obtain the identity. Concavity makes each summand nonnegative. For the equality condition, let $a_i = s_i + r_i$ and use

$$g_i(q) = s_i q - a_i (q - \theta_i)_+.$$

The conditional gap is

$$a_i (\mathbb{E}[(q_i - \theta_i)_+ \mid z, y] - (\mathbb{E}[q_i \mid z, y] - \theta_i)_+).$$

If all posterior values are weakly on one side of the threshold, this is zero. If positive mass lies strictly on both sides and the mean is at least the threshold, the expression in parentheses equals $\mathbb{E}[(\theta_i - q_i)_+ \mid z, y] > 0$. If the mean is at most the threshold, it equals $\mathbb{E}[(q_i - \theta_i)_+ \mid z, y] > 0$. Positive joint support makes these statements apply to every member of each fiber, not merely an almost-sure subset. The interval criterion in Theorem 5.1 gives the equivalent common-optimum condition. $\square$

The theorem concerns selecting *any* optimum. A decoder need not reproduce an arbitrarily chosen canonical full-information policy at a tie. In particular, equality of such a policy is sufficient but can be unnecessarily restrictive for zero regret.

Example 5.3 (A strict regret gap and a tie-policy distinction). Two equally likely contexts have one uninformative signal and one artifact, with $q(L) = 1/4$ and $q(H) = 3/4$. For $s = r = 1$, full-information risk is $1/4$. Merging the contexts yields $\bar{q} = 1/2$ and risk $1/2$, so representation regret is $1/4$.

For $s = 1, r = 3$, the threshold becomes $3/4$. Retaining the artifact is optimal in both contexts, giving one-code zero regret and risk $1/2$. A prescribed “reopen on ties” policy instead retains at L and reopens at H , requiring two codes to reproduce that policy. The extra code preserves the tie convention rather than improving Bayes risk.

With posteriors below, equal to, and above one threshold, the middle context is compatible with either extreme and the extremes conflict. Thus common-optimum compatibility at fixed costs need not be transitive. This is the stochastic realization of the optional-edit obstruction in Section 7.

5.4 A cost-universal sufficient signature

Let

$$Q_{\text{imp}}(w) = (q_i(w, y))_{i \in \mathcal{I}, y \in \mathcal{Y}}.$$

A statistic $T : \Omega \rightarrow \mathcal{Z}$ is *cost-universal repair-sufficient* if it is chosen independently of the cost vector and, for every vector of positive (s_i, r_i) , admits a cost-aware decoder optimal at every (w, y) . Artifact identities and signal meanings are aligned across contexts.

Theorem 5.4 (Coarsest universal additive statistic). *Under the preceding finite, common-support, additive, unrestricted-action assumptions, T is cost-universal repair-sufficient if and only if it determines Q_{imp} . Equality of that signature is therefore the unique coarsest such partition, up to relabeling.*

Proof. If the statistic determines all posterior entries, the decoder applies Theorem 5.1 for the supplied costs. Conversely, suppose two contexts with the same statistic differ at entry (i, y) , with values $a < b$. Choose $\theta \in (a, b)$ and set $s_i = 1 - \theta$, $r_i = \theta$. These are positive and give threshold θ . Fix arbitrary positive costs for other coordinates. The two contexts have opposite unique optimal actions at coordinate i for the same observed signal, so one retained code cannot be sufficient. Every sufficient partition must therefore refine signature equality, which is itself sufficient. $\square$

There may be no strict compression: every context can have a different signature. This is a task-specific sufficiency result, not a likelihood-sufficiency theorem or an efficient finite-precision encoding scheme. Restricting the cost family, allowing approximation, or admitting context-dependent missing signals can change the required distinctions.

5.5 Nested information and coupled choices

An oracle comparison must actually contain the information available to the less informed observer. For the present model the nested information states are (W, D, Y) , (W, Y) , and $(t(W), Y)$. Each richer observer can imitate the poorer policy, so

$$0 = B_{\text{oracle}} \leq B_{\text{full}} \leq B_t, \quad B_t - B_{\text{oracle}} = (B_{\text{full}} - B_{\text{oracle}}) + (B_t - B_{\text{full}}).$$

The oracle has zero loss because it knows all impacts and all subsets are feasible. Observing D alone need not be more informative than observing (W, Y) . The displayed decomposition separates two information gaps under one fixed model; it is not a unique attribution of every implementation, provenance, or evaluator error.

Now impose a known nonempty feasible family $\mathcal{F} \subseteq 2^{\mathcal{I}}$. Expected loss at posterior vector q becomes

$$\sum_i s_i q_i - \sum_{i \in S} [(s_i + r_i) q_i - r_i], \quad S \in \mathcal{F}. \quad (8)$$

Coordinatewise thresholding may violate $\mathcal{F}$, and optimal actions may no longer form a Boolean interval. However, the marginal posterior vector still determines the additive objective. It suffices for optimization when the same feasible family is known. Correlations are not required solely because decisions are coupled. They become relevant if the loss or constraints depend on joint outcomes.

If hidden contexts have different feasible families, the posterior vector alone need not suffice: identical impact probabilities can accompany different permissions or predecessor requirements. An equivalent description of that feasible family, or other adequate decision signature, must then be retained.

The common-feasible-action requirement across a compressed cell must also be defined. For a fixed union/intersection-closed family and one scalar nonnegative cost path, Appendix A proves the scoped $n + 1$ bound on canonical optimal sets. That bound does not solve the general continuous-cost, uncertain-constraint representation problem.

6 Maintaining a correction-sufficient state online

One-shot sufficiency does not ensure that a representation can be updated without rereading the reference state. We now add an explicit sequential contract to Section 3. Each event has a total deterministic transition $\tau_e : X \rightarrow X$. The output for (x, e) is selected before the transition, and the transition is unaffected by that output. All event sequences and all initial states are allowed.

The encoder observes the initial reference state and stores $z_0 = c(x_0)$. Thereafter a deterministic controller sees only its symbol and the current fully observed event:

$$\Delta_t = d(z_t, e_t), \quad z_{t+1} = g(z_t, e_t), \quad x_{t+1} = \tau_{e_t}(x_t).$$

The defining restriction is

$$z_t = c(x_t) \quad \text{at every time.} \tag{9}$$

No clock, past-event log, reference-state reread, additional receipt observation, or future event is freely available. The model counts occupied symbols, with transitions and decoder in a shared codebook. It concerns a maintainable abstraction of the current reference environment. In the running example this is the changing evidence-root set. The full operational ledger from Section 2 can change as a consequence of a repair and is not assumed to obey this exogenous-transition model.

General history-dependent controllers, stochastic or partial observations, action-dependent environments, and overlapping state covers are different problems. The minimum below is specifically over fixed-current-state encodings satisfying (9).

Theorem 6.1 (Correction-compatible transition congruence). *An encoder satisfies the sequential contract for some decoder and updater if and only if*

1. *every encoder cell is independent in the correction-conflict graph G ; and*
2. *$c(x) = c(y)$ implies $c(\tau_e(x)) = c(\tau_e(y))$ for every event e .*

Consequently

$$\chi(G) \leq K_{\text{online}} := \min\{|\mathcal{P}| : \mathcal{P} \text{ is transition-stable with } G\text{-independent blocks}\} \leq |X|.$$

Proof. Every state may initialize the process. At its first event, all members of one cell receive the same output, so Theorem 3.4 requires independence. The same initial symbol and event also produce the same next symbol; condition (9) forces equality of the successor encodings. This proves necessity.

Conversely, choose $d(z, e) = \bigcup_{x \in X_z} M_e(x)$, valid by independence. Set $g(z, e) = c(\tau_e(x))$ for any $x \in X_z$. Transition stability makes this well-defined. Induction on time proves (9) and the validity of every output. The discrete partition is always feasible, so the finite minimum exists and the stated bounds follow. $\square$

The condition is an ordinary transition-congruence requirement with compatible set-valued outputs. The kernel already notes the extra update obligation; the subsequent sequential note supplies the explicit characterization and counterexamples [2, 5]. Deterministic automaton minimization provides the established future-output mechanism [13].

6.1 A smallest strict gap

With $X = \{R, O, F\}$, one event, and one edit atom a , set

State	Required M	Permitted U	Successor
R	$\{a\}$	$\{a\}$	O
O	$\emptyset$	$\{a\}$	F
F	$\emptyset$	$\emptyset$	R

The letters indicate required, optional, and forbidden output. The graph has only the edge RF .

Proposition 6.2 (Smallest strict one-shot/online separation). *This system has $K_{\text{one}} = 2$ and $K_{\text{online}} = 3$. No system satisfying the same contract with fewer than three states has a strict gap, regardless of its finite event and edit universes. One event and one edit atom suffice, and zero edit atoms cannot give a gap.*

Proof. The only feasible two-block partitions are $\{R, O\} \mid \{F\}$ and $\{R\} \mid \{O, F\}$. The first merges R, O but separates their successors O, F ; the second merges O, F but separates their successors F, R . Neither is stable. The discrete partition is stable. The graph edge proves that two and not one one-shot symbols suffice.

For two states the only partitions are indiscrete and discrete. If the indiscrete partition is one-shot feasible, it is stable under every total transition. Otherwise both counts are two, since the discrete partition is stable. One state is immediate. With no edit atoms every interval has the sole patch $\emptyset$, so the one-cell partition always works. $\square$

Proposition 6.3 (An unbounded state-count separation). *For each $n \geq 2$, a one-event n -cycle with output necessarily $\{a\}$ at state 0 and necessarily $\emptyset$ elsewhere has $K_{\text{one}} = 2$ and $K_{\text{online}} = n$.*

Proof. The conflict graph is a star with center 0, giving two one-shot codes. For any distinct states i, j , some common number of transitions sends i to 0 and leaves j elsewhere. A stable encoding merging i, j would also merge these future states, whose required outputs conflict. Every pair must be separated, and the discrete encoding works. $\square$

This is classical future-output distinguishability expressed in correction terms. Section 4's four-state evidence example supplies a less abstract instance: three one-shot symbols and four maintainable ones. Neither count is a claim about token efficiency in an actual research assistant.

7 Minimum size, coarsest quotients, and proposed merges

A minimum-cardinality valid partition uses as few blocks as possible. A *coarsest* valid partition must be coarser than *every* valid partition under refinement. A finite minimum always exists here, whereas a coarsest element need not.

Keep the intervals for R, O, F but make the transition the identity. Every partition is now stable. Both $\{R, O\} \mid \{F\}$ and $\{R\} \mid \{O, F\}$ are valid minimum two-block partitions. Any partition coarser than both merges all three states by transitivity and contains the forbidden pair R, F . Thus no coarsest valid partition exists. Optional outputs produce nontransitive pair compatibility; the same obstruction occurs at a fixed Bayes-cost tie.

7.1 Future compatibility is necessary but insufficient for a merge

For a finite event word w , write $\tau_\varepsilon = \text{id}$ and $\tau_{we} = \tau_e \circ \tau_w$. Define $x C_\infty y$ when, for every word w and next event e , the intervals at $\tau_w(x)$ and $\tau_w(y)$ have a common patch. This relation is reflexive, symmetric, and forward stable. Every online-valid equivalence is contained in it.

Pairwise future compatibility does not alone allow a fixed-state merge. Set $R \rightarrow O \rightarrow F \rightarrow F$ with the same intervals. States R, O remain pairwise compatible at each common future time: (R, O) , then (O, F) , then (F, F) . Merging R, O nevertheless forces merging O, F after one event, and equivalence transitivity then merges R, F , which conflict. The valid coarsest partition in this example is $\{R\} \mid \{O, F\}$. Thus nontransitivity of C_∞ alone does not characterize whether a coarsest valid quotient exists.

7.2 A finite certificate for proposed identifications

Theorem 7.1 (Least congruence closure). *Start with any proposed identifications and close them under equivalence. Repeatedly identify $\tau_e(x)$ and $\tau_e(y)$ whenever x, y are identified, again closing under equivalence. The procedure terminates at the least transition congruence R^* containing the proposal. An online-valid encoding containing every proposed identification exists if and only if every R^* class is G -independent.*

Proof. Every forced identification belongs to every congruence containing the proposal, inductively by equivalence closure and transition stability. Each actual merge reduces the number of classes, so at most $|X| - 1$ occur. At termination no required successor identification remains, giving a congruence contained in every candidate. If it contains a conflict, any coarser relation preserves that conflict and is invalid. If it contains none, Theorem 6.1 supplies a decoder and updater. $\square$

Define $x R_{\text{merge}} y$ when the least congruence generated by the pair (x, y) contains no conflict.

Corollary 7.2 (Exactly when a coarsest quotient exists). *A coarsest online-valid partition exists if and only if R_{merge} is transitive. If it exists, its blocks are the equivalence classes of R_{merge} .*

Proof. By Theorem 7.1, R_{merge} is exactly the union, as ordered pairs, of all online-valid equivalence relations. It is reflexive and symmetric. A witnessing valid congruence for a pair also witnesses each successor pair, so the union is forward stable. No conflict belongs to the union. If it is transitive, it is an equivalence and a congruence with independent classes; the interval property and Theorem 6.1 make it valid. Its union characterization makes it coarser than every valid partition. Conversely, a coarsest valid equivalence contains every valid pair relation and is itself included in their union, so it equals that union and is transitive. $\square$

These statements give finite tests for a proposed merge and for the existence of a coarsest quotient. They do not supply a polynomial algorithm for minimum-cardinality encoding when a coarsest quotient does not exist. Exhaustive enumeration of all partitions is a separate finite optimization method. Conservative complexity and exact model boundaries appear in Appendix B.

7.3 Fixed signatures recover ordinary refinement

Let $h : X \rightarrow H$ be any fixed signature whose exact value must be recoverable. Define

$$x \sim_h y \iff h(\tau_w(x)) = h(\tau_w(y)) \text{ for every finite word } w.$$

Proposition 7.3 (Coarsest maintainable exact signature). *The relation $\sim_h$ is the unique coarsest transition congruence refining equality of h . It is obtained by starting with h -equality and repeatedly splitting blocks by the tuple of successor blocks until stable, after at most $|X| - 1$ strict refinements.*

Proof. Equality of every future signature is an equivalence, refines current h -equality using the empty word, and is preserved by every transition using the corresponding prefixed word. Any stable equivalence preserving h preserves its value after every word by induction and therefore refines $\sim_h$. In the iterative construction, every such congruence refines every intermediate partition. Termination follows because each strict refinement increases the finite block count. The final stable partition itself preserves h , so it is the stated coarsest relation. $\square$

When every acceptable patch is a singleton, take $h(x)$ to be the table of unique outputs over current events. This is the familiar deterministic output-machine case [13]. For the universal additive model in Theorem 5.4, take $h = Q_{\text{imp}}$ if an additional sequential model supplies the exogenous transitions and fixed-state contract above. Equality of the complete future posterior-impact behavior is then the coarsest maintainable universal statistic. This conditional implication does not provide a transition law for the historical RR manuscript. Reproducing a canonically selected action from a set of tied optima can impose a stricter task than preserving the ability to select any optimum.

8 Relation to established work

The mathematical mechanisms used here have established antecedents. The common-patch criterion is an elementary intersection calculation for Boolean intervals. Its graph formulation exploits the resulting pairwise-to-global compatibility property. Characteristic graphs and graph-based restrictions on coding are established in zero-error computation; Charpenay, Le Treust, and Roumy provide a contemporary formulation with side information [14]. Our quantity is the number of fixed, one-shot codes for an explicit finite interval model. It is not their asymptotic rate or an independently established improvement to coding theory.

Grounded support belongs beside truth maintenance and provenance. De Kleer’s assumption-based truth-maintenance system records alternative assumption environments supporting a belief [10]. The positive Horn model in Section 4 captures only a restricted part of that problem: grounded conjunction, alternative routes, and withdrawal. It omits the richer treatment of assumptions and inconsistencies in an ATMS. Provenance semirings give formal accounts of how source contributions combine [11]; the W3C PROV data model distinguishes entities, activities, agents, and derivation [12]. These connections motivate explicit support and provenance records. They do not make a recorded claim true, grant an agent authority, or establish that a proposed change was performed.

The stochastic results use conditional expectation, additive Bayes loss, threshold decisions, and concavity. Their significance here is the explicit bridge from a fixed-cost optimal-action interval to the robust common-action criterion, with a separate cost-universal signature. The sequential results use equivalence relations stable under transitions. Partition refinement and finite-state minimization are established techniques; Hopcroft’s automaton-minimization report is a primary reference [13]. An automaton with a fixed output label has a different specification from a controller allowed several acceptable outputs. That distinction explains why a coarsest valid quotient is automatic in the fixed-signature setting but can fail in the interval setting. No new general automaton-minimization algorithm is claimed.

The human–AI motivation also has narrower empirical predecessors than a universal claim about collaboration. Pronin, Lin, and Ross report a self/other asymmetry in perceived susceptibility to bias [6]. Fleming and Daw formalize informative but imperfect metacognitive computation [7]. Vaccaro, Almaatouq, and Malone distinguish improvement over a human alone from improvement over the better standalone actor, and find heterogeneous outcomes in their reviewed human–AI studies [8]. Buçinca, Malaya, and Gajos study cognitive forcing functions in a particular AI-assisted decision task [9]. These works motivate testable mechanisms and appropriate comparators. None validates ETI’s intervention or warrants an inference about human or machine consciousness.

This positioning is a targeted comparison of primary sources, not a systematic priority review. The accompanying reference record distinguishes full documents available from passages actually inspected; the Pronin comparison uses the publisher abstract and metadata. Mathematical correctness within the stated model, historical originality, implementation fidelity, and practical advantage remain separate questions.

9 Evidence and reproducibility

9.1 What the different forms of evidence establish

Four evidence levels should be kept distinct. A proof establishes a proposition under its assumptions. A finite implementation check compares code with a declared reference on enumerated inputs. An instrument-validity study tests whether inputs, conditions, graders, and measurements represent the intended task. A comparative outcome study tests whether an intervention improves specified outcomes at a measured cost. Evidence at an earlier level does not establish a later level.

The proofs in this paper supply the general arguments for the scoped mathematical claims. Fresh, independently written finite checkers corroborate them without importing the earlier ETI implementations. Their domains deliberately include ties, alternative support, ungrounded cycles, conflicting requirements, and counterexamples produced by dropping assumptions. Table 1 records the new checks. Exact integers and rational arithmetic avoid numerical decisions at threshold ties.

Additional fixtures check the five-cycle code count, the three-state online gap, incomparable optimal partitions, future pair compatibility without a valid merge, and the four-state withdrawal example. Cyclic examples are checked for two through eight states. Decision fixtures check signal-conditioned mixing weights, the difference between preserving any optimum and preserving a chosen tie policy, and failures of scalar nesting without its closure, sign, or extremal-selection assumptions. These are fresh reconstructions, not a recovery of missing historical validation code. They do not exhaust unbounded systems, uncertain probabilities, natural-language state construction, or arbitrary repair constraints.

9.2 Historical finite evidence and unresolved provenance

The earlier kernel reference suite has a larger but different finite domain [2, 4]. It records 19,683 ordered triples of valid intervals on three edit atoms, 98,415 partition/decoder checks, and 19,683 exact-coloring comparisons. Its support corpus comprises 4,096 subsets of twelve rule templates crossed with eight evidence subsets, giving 32,768 support states. Three single-evidence toggles per state give 98,304 directed transitions, including 49,152 removals. Eight authored adversarial fixtures supplement that corpus. A recorded September 5 audit reproduced the historical result files; this synthesis preserves that historical attribution rather than describing it as a new execution.

New check domain	Accounting unit	Scope
Common correction	729 triples; 3,645 partitions	Three states, two edit atoms; all interval triples and partitions.
Information refinement	5,103 subsets	All nonempty subsets within those triples.
Sequential characterizations	2,064 systems; 6,312 encodings	Up to three states with one event, and two states with two events; one edit atom.
Grounded support	1,024 states; 1,024 removals	Two claims, two evidence roots, eight rule templates.
Single-artifact Bayes loss	9,000 weighted cases	Three worlds; posterior and positive-cost grids; exact threshold ties.
Joint impact distributions	720 cases	Two artifacts; explicit joint distributions and positive costs.
Scalar closure paths	13,974 systems; 47,658 parameter positions	Up to three artifacts; graph-closed families, exact breakpoint and tie checks.

Table 1: Fresh synthesis checks. Counts describe different, overlapping enumeration units and are not summed into a number of independent experiments. All listed checks passed.

Within those 49,152 removal scenarios, exact grounded-support maintenance recorded no missed or unnecessary claim reopens against the declared finite reference. A descendant-traversal baseline made 24,096 unnecessary claim-reopening decisions. A decision count is not a scenario count. This comparison illustrates what the reference semantics require; it is not a comparison against an equally resourced ATMS, production agent, or competitive support-maintenance implementation. The finite note is a condensation of the kernel, not independent corroboration of it.

The RR-ETI manuscript reports a historical 512-policy validator [3]. That code has not been recovered for this synthesis, so those historical checks are not represented as freshly replayed. Its four-world example has four distinct posterior-impact signatures and therefore does not itself demonstrate compression by that statistic. The reconstructed proofs and new rational checkers provide a separate, explicitly scoped basis for the statements made here.

Study 14 has a development package with 28 passing test methods covering such mechanisms as all-stage metric handling, structured support construction, and condition-input parity. Those methods are development controls. They neither reconstruct the older prototype’s test total nor constitute a held-out comparison. Earlier prototype test success did not establish intended propagation and authority behavior; that limitation is not repaired by relabeling the kernel’s proof or finite tests as system efficacy.

9.3 Reproduction and review boundaries

The companion package contains the editable L^AT_EX source, exact source identities, independent finite checkers, their execution records, a claim ledger, and review notes. The deterministic and rational checkers use the Python standard library. They operate on generated finite objects, make no model calls, and do not read hidden Study 14 cases. Instructions distinguish recomputation of the new checks from the historical results summarized above.

AI review roles separately examined the deterministic theory, the stochastic theory, and the conceptual and reference claims, followed by review of the integrated manuscript. This role separation is useful for exposing inconsistencies but is not blinded human review, external peer review, formal proof-assistant verification, or statistically independent replication. The resulting evidence supports the stated conditional claims; it cannot guarantee that no error remains.

10 A falsifiable empirical program

10.1 The missing bridge is construction and use of state

The theorem asks what information would suffice if its interpretation and transitions were correct. The empirical question is whether a system can construct, maintain, and use that information from the same permitted material available to its comparators, with acceptable error and cost. A reference state supplied by the task author is useful for a ceiling demonstration. It cannot silently serve as the deployed system's state constructor.

The operational representation should record the origin of each evidence item, support rule, status assignment, authority constraint, and closure obligation. A claim of completion needs an outcome record appropriate to the task; a planned action or fluent statement of success is not such a record. Construction errors, unsupported authority assignments, missed updates, and false closure decisions should remain measurable even when a mathematical decoder behaves correctly on its supplied input.

In the existing Study 14 development work, the constructor parses authored structured support statements while excluding hidden grading references from its input interface. This establishes a mechanical separation at that interface and exercises a limited support model. It does not establish independent natural-language construction, calibrated source authority, or a finished scheduler. Those remain empirical and engineering obligations.

10.2 Preserving the actual comparison

The intended Study 14 conditions are:

A: full context.

All permitted raw context, with the target and current task obligations pinned.

B: context with shared status information.

The same raw text and order as A, plus the same status and provenance information used by D, without dependency-governed scheduling.

C: ordinary semantic retrieval.

A competitive, frozen retriever over permitted raw evidence, without privileged authority or dependency labels in its index.

D: state-governed scheduling.

Working context selected and updated through independently constructed operational state, with the stated dependency and revision obligations.

B is needed to distinguish an advantage from status labels from an advantage from scheduling with those labels. The full target, completion criteria, correction message, tool contract, and output

format are pinned in every arm. A and B must preserve source content and order and fit within the common window; matching source identifiers alone is insufficient. B and D share the cost of constructing the statuses they share. A checklist comparison, human-only condition, or different ablation would require its own explicit design; it is not a renamed version of these four arms.

C’s competitive retriever and D’s full operational constructor and scheduler are not yet frozen. Hidden answer keys, permission keys, and affected-conclusion reference sets must not enter D’s construction inputs or an equivalent proxy label. A source-interface check cannot by itself exclude all such informational advantages. Independent construction assessment and condition validation therefore precede comparative interpretation.

10.3 Outcomes, costs, and disconfirmation

The base case is the paired inferential unit. Stages within a case and variants of the same case should not be counted as independent observations. Primary outcomes include any-stage unauthorized premise reliance, behavioral correction after a relevant update, and any-stage false completion. Correction precision, valid completion, retained accuracy, and overblocking are necessary guardrails. A trace that records a reopen is not evidence that the downstream response was corrected.

Development cases, pilot cases, and held-out evaluation cases have distinct roles. Condition labels should be masked in grading when feasible. The task distribution, scoring rules, disagreements, primary contrasts, multiplicity treatment, practically meaningful margins, and sample-size rationale should be fixed before confirmatory interpretation. A nonsignificant difference does not establish practical equivalence. No effect size, power, or human–AI performance result is inferred from the finite check counts.

Cost accounting includes state construction, maintenance, task execution, adjudication, human effort, latency, tokens, and actual charges where available. Unknown cost fields remain unknown. One-time engineering costs and marginal execution costs should be reported separately, and relevant shared construction costs allocated consistently. The coding bounds in Sections 3 and 6 are not end-to-end resource bounds.

The intervention would be weakened or disconfirmed for a stated task distribution if a simpler, equally informed method were practically equivalent at lower cost; if fewer stale conclusions were purchased through unacceptable overblocking; if an apparent advantage disappeared when hidden reference information was removed; or if the constructor could not reliably identify the required dependencies and authority distinctions. A gain over a weak retriever would not establish a gain over a competitive alternative. A successful result in one task class would still require a declared transfer boundary.

11 Contribution, limitations, and accountability

This synthesis supplies a common vocabulary, connected mathematical specifications, explicit proofs, counterexamples, and an evaluation contract for correction under partial information. The central connection is precise: complete interval-valued acceptable actions admit a common-action test and a conflict graph; unrestricted additive fixed-cost optimal actions instantiate the same form; persistent encoding of a changing reference state adds transition congruence. Those connections organize the research without identifying distinct notions of sufficiency with one another.

The restrictions are substantial. The robust model assumes finite states and compatible edit atoms with all intermediate patches allowed. The stochastic signature result assumes a specified proba-

bilistic model, common positive signal support, and common positive additive costs, and all repair subsets feasible. The online characterization assumes a deterministic, exogenous transition system and a fixed encoding of current reference state, with no extra history or clock. Natural-language evidence interpretation, conflicting authorities, uncertain model parameters, action-dependent dynamics, and empirical system reliability require further treatment. Several limits are witnessed by counterexamples, not merely listed as possible concerns.

The conceptual commitment is to make research correction inspectable: preserve the distinction between an observation and an inference, between support and permission, and between an action description and its outcome. That commitment may improve a workflow only if the resulting records are constructed and used well enough to justify their costs. It is a research hypothesis with a finite mathematical specification, not a general theorem that humans and AI together discover more truth.

Contribution and AI assistance. Zackary Loevseth directed the research problem, requested the consolidation, and supplied the source corpus and intended use. AI systems materially assisted the formalization, proofs, counterexamples, reference code, source reconciliation, literature checking, and writing. This paper does not assert that the project owner independently derived or personally verified every proof, reference, or line of code. The local review and reproducibility records describe what was actually checked. External peer review, journal acceptance, and empirical superiority are not claimed. The synthesis is a new local manuscript version; original source documents and historical releases remain separate records.

A A fixed scalar closure path

One useful specialization of the RR program is a scalar tradeoff on a fixed closure family. Its assumptions should not be confused with the cost-universal statistic in Theorem 5.4.

Let I contain n artifacts, and let $\mathcal{F} \subseteq 2^I$ be a nonempty family closed under union and intersection. For fixed weights $w_i \in \mathbb{R}$, costs $c_i \geq 0$, and a scalar $\lambda \geq 0$, maximize

$$f_\lambda(S) = \sum_{i \in S} (w_i - \lambda c_i), \quad S \in \mathcal{F}.$$

A graph implication constraint (selecting an artifact requires selecting specified others) gives one such family. A cardinality budget need not do so. All weights, costs, constraints, and any state graph are fixed while λ varies.

Theorem A.1 (Nested extremal optimizers). *At each λ , the maximizing sets have a unique inclusion-least member and a unique inclusion-greatest member. If either extremal convention is used consistently, its selected set decreases by inclusion as λ increases. It takes at most $n + 1$ distinct values; this bound is attained by some instances.*

Proof. Modularity gives

$$f_\lambda(P) + f_\lambda(Q) = f_\lambda(P \cup Q) + f_\lambda(P \cap Q).$$

If P, Q maximize the objective, both sets on the right are feasible and neither exceeds the maximum. Therefore both maximize it. Finite intersections and unions of all maximizers supply the two unique extrema.

Let $\lambda_1 < \lambda_2$, let P maximize at λ_1 , let Q maximize at λ_2 , and set $D = Q \setminus P$. Feasibility of $P \cup Q$ gives $f_{\lambda_1}(D) \leq 0$; feasibility of $P \cap Q$ gives $f_{\lambda_2}(D) \geq 0$. Here the modular expression defines $f_\lambda(D)$

even if D itself is not feasible. Since $c_i \geq 0$,

$$0 \geq f_{\lambda_1}(D) \geq f_{\lambda_2}(D) \geq 0.$$

Both contributions are zero. Consequently $P \cap Q$ also maximizes at λ_2 , and $P \cup Q$ also maximizes at λ_1 . If Q is the least optimizer at λ_2 , then $Q = P \cap Q$, so $Q \subseteq P$. If P is the greatest optimizer at λ_1 , then $P = P \cup Q$, giving the same conclusion. Applying the appropriate argument to the consistently chosen extrema proves nesting. A strict decrease removes at least one of the n artifacts permanently, giving at most $n + 1$ distinct sets. For sharpness, take $\mathcal{F} = 2^{\{1, \dots, n\}}$, $c_i = 1$, and $w_i = i$. Passing the distinct thresholds $1, \dots, n$ removes the artifacts one at a time, producing $n + 1$ sets under either extremal convention. $\square$

Three boundaries matter. If $\mathcal{F} = \{\emptyset, \{a\}, \{b\}\}$, $w = (3, 2)$ and $c = (3, 1)$, the unique optimum changes from $\{a\}$ at $\lambda = 0$ to $\{b\}$ at $\lambda = 3/4$, violating nesting; the family lacks union closure. With negative costs, an artifact may enter as λ rises. Finally, even a fixed arbitrary ordering for ties is insufficient: on $2^{\{a, b\}}$ with $w_a = c_a = 1$ and $w_b = c_b = 0$, ordering $\{a\}, \{b\}, \emptyset, \{a, b\}$ selects $\{a\}$ below 1 and $\{b\}$ above 1. Extremal selection avoids this problem. The bound counts the chosen scalar path, not every optimizer, every feasible set, a multidimensional Pareto frontier, or a changing graph.

B Finite algorithms and explicit input costs

For a finite explicit model, the proofs give direct procedures. A common-patch check forms the required union and permitted intersection. Enumerating all partitions and retaining those whose blocks pass that check computes the exact one-shot minimum. Adding the transition-congruence check computes the minimum number of online codes in the fixed-encoding class. These exhaustive optimization procedures grow rapidly with the number of states; their availability is not a claim of scalable optimization.

For a proposed pair merge, begin with the equivalence generated by that pair. Repeatedly inspect every current equivalent pair (x, y) and every event e . If their successors lie in different blocks, merge those blocks and restore equivalence. Each successful pass reduces the number of blocks, so at most $n - 1$ successful passes occur. Once no merge is needed, test every resulting block for correction feasibility. Theorem 7.1 proves that failure rules out every valid congruence containing the proposed pair, and success supplies one. Applying this test to every pair determines the relation used in Corollary 7.2; transitivity of that relation decides whether a coarsest valid quotient exists. It does not solve minimum-cardinality encoding in all cases.

For clarity, consider a deliberately simple implementation with $n = |X|$, $m = |\mathcal{E}|$, and $v = |\mathcal{V}|$, where $m \geq 1$. Transitions are an explicit $n \times m$ array of state indices, and interval endpoints are explicit v -bit sets. A parent-pointer representation with worst-case $O(n)$ root lookup and repeated full pair scans gives the conservative word-operation bound

$$O(mn^4 + mn^2v)$$

for one proposed merge and its final pairwise interval-conflict check. The two terms bound congruence closure and explicit conflict inspection. Performing all pair tests and then a direct transitivity test gives

$$O(mn^6 + mn^4v + n^3).$$

These are upper bounds for a naive explicit-input procedure, not optimized complexity claims. Input plus working storage can be bounded by

$$O\left((n + mn)\lceil \log_2 \max(2, n) \rceil + nmv + n^2\right) \text{ bits,}$$

including the all-pair relation. This accounts for the explicit state model, which must already have been constructed. It does not account for learning or discovering that model from text. More efficient partition-refinement methods are available for fixed-output models [13]; their assumptions must be checked before transferring a complexity result to interval outputs.

C Source lineage and corrected statements

This manuscript synthesizes five source documents. Their exact bytes and hashes are recorded in the companion source-identity file. The table assigns a role to each source; it does not declare all claims in an older document verified or superseded.

Source	Title and version	Role in this synthesis
S1	<i>Epistemic Transition Integrity: Human–AI Research, Blind Spots, and Accountable Discovery</i> ; ETI-FOUNDATION-2026-09-04.1	Motivation, typed architecture, reciprocal correction hypothesis, and research boundaries.
S2	<i>ETI: A Mathematical Kernel—Sufficient information for justified correction</i> ; ETI-MATH-2026-09-04.1	Finite correction intervals, coloring, grounded support, and historical reference-suite evidence.
S3	<i>Revision-Robust Epistemic Transition Integrity: Technical Core, Operational Runtime, and Current Evidence</i> ; v2.0, 13 August 2026	Conditional repair loss, posterior sufficiency, and fixed scalar closure specialization, reconstructed with explicit assumptions.
S4	<i>Correction-Sufficient State for Revision Under Partial Information</i> ; ETI-FINITE-0.1.0-rc1, 7 September 2026	Compact finite statement and clarified claim and release boundaries; shares the kernel lineage.
S5	<i>When a correction-sufficient state can be maintained online</i> ; sequential note v2, 7 September 2026	Transition congruence, strict code-count gaps, coarsest-quotient boundaries, and proposed-merge test.

Several statements are deliberately sharpened in the synthesis. Preserving Bayes risk means preserving some optimal action, not necessarily reproducing an arbitrarily imposed tie-breaking policy. An information oracle is explicitly nested by giving it (W, D, Y) ; an event variable alone need not contain the information in (W, Y) . Coupled action choices invalidate an unrestricted coordinatewise optimizer, but posterior marginals still determine additive expected loss over a known common feasible family. Correlations become relevant when the loss or feasibility actually uses joint information. Scalar closure nesting requires a fixed lattice of feasible sets, nonnegative cost slopes, and a coherent extremal optimizer. Online congruence characterizes a fixed encoding of current reference state, not every possible history-dependent controller.

The withdrawal example also distinguishes reference-environment dynamics from changes made to an operational ledger by a chosen action. The present transition theorem applies to the former under its

exogenous-transition assumption. An action-dependent ledger requires a correspondingly different dynamic model. The manuscript’s new four-state illustration connects already specified support and quotient mechanisms; it is an explanatory consequence, not an established priority claim.

D Regimes at a glance

Regime	What is preserved	Additional conditions or exclusions
Robust finite correction	One patch is acceptable at every reference state in a cell for the observed event.	Nonempty cells; compatible edits; complete intervals $[M_e(x), U_e(x)]$; no probability averaging.
Fixed-cost Bayes repair	Some action remains conditionally optimal at each supported (w, y) .	Known additive costs and model; all subsets feasible; derived optimal-action intervals; ties may admit different policies with equal risk.
Cost-universal Bayes repair	Optimality is retained for every positive cost vector, with decoder access to that vector.	One statistic works for all costs; common positive signal support; it retains the complete posterior-impact signature.
Online fixed encoding	The retained code equals the same encoding of the current reference state at every step.	Exogenous deterministic total transitions; congruence as well as valid code cells; no extra hidden memory.
Empirical research system	The required distinctions are constructed and used from permitted sources with acceptable error and cost.	Not established by these theorems; requires instrument validation and competitive outcome comparisons.

References

- [1] ETI research corpus. *Epistemic Transition Integrity: Human–AI Research, Blind Spots, and Accountable Discovery*. ETI-FOUNDATION-2026-09-04.1. Local foundational reference, September 2026. Source S1 in Appendix C.
- [2] ETI research corpus. *ETI: A Mathematical Kernel—Sufficient information for justified correction*. ETI-MATH-2026-09-04.1. Local manuscript and reference implementation, September 2026. Source S2.
- [3] ETI research corpus. *Revision-Robust Epistemic Transition Integrity: Technical Core, Operational Runtime, and Current Evidence*. Version 2.0, 13 August 2026. Local technical manuscript. Source S3.
- [4] ETI research corpus. *Correction-Sufficient State for Revision Under Partial Information*. ETI-FINITE-0.1.0-rc1, 7 September 2026. Local finite note. Source S4.
- [5] ETI research corpus. *When a correction-sufficient state can be maintained online*. Version 2, 7 September 2026. Local sequential-state note. Source S5.
- [6] E. Pronin, D. Y. Lin, and L. Ross. The bias blind spot: Perceptions of bias in self versus others. *Personality and Social Psychology Bulletin*, 28(3):369–381, 2002. <https://doi.org/10.1177/0146167202286008>.
- [7] S. M. Fleming and N. D. Daw. Self-evaluation of decision-making: A general Bayesian framework for metacognitive computation. *Psychological Review*, 124(1):91–114, 2017. <https://doi.org/10.1037/rev0000045>.
- [8] M. Vaccaro, A. Almaatouq, and T. Malone. When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8:2293–2303, 2024. <https://doi.org/10.1038/s41562-024-02024-1>.
- [9] Z. Bućinca, M. B. Malaya, and K. Z. Gajos. To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1), Article 188, 1–21, 2021. <https://doi.org/10.1145/3449287>.
- [10] J. de Kleer. An assumption-based TMS. *Artificial Intelligence*, 28(2):127–162, 1986. [https://doi.org/10.1016/0004-3702\(86\)90080-9](https://doi.org/10.1016/0004-3702(86)90080-9).
- [11] T. J. Green, G. Karvounarakis, and V. Tannen. Provenance semirings. In *Proceedings of the 26th ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems (PODS)*, pages 31–40, 2007. <https://doi.org/10.1145/1265530.1265535>.
- [12] L. Moreau and P. Missier, editors. *PROV-DM: The PROV Data Model*. W3C Recommendation, 30 April 2013. <https://www.w3.org/TR/2013/REC-prov-dm-20130430/>.
- [13] J. Hopcroft. *An $n \log n$ algorithm for minimizing states in a finite automaton*. Technical Report STAN-CS-71-190, Stanford University, January 1971. <https://www.cs.cmu.edu/~cdm/resources/Hopcroft1971.pdf>.
- [14] N. Charpenay, M. Le Treust, and A. Roumy. Side information design in zero-error coding for computing. *Entropy*, 26(4):338, 2024. <https://doi.org/10.3390/e26040338>.